

Practical No. 9

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Aim : Create DFD's (Data Flow Diagram), Decision tables and E-R (Entity-relationship) diagram.

DFDs, Decision Table and E-R Diagram

Sample Project: Online Food Ordering System

Objective: To understand and prepare DFDs, a decision table, and an E-R diagram for a simple software project.

Project Description: The Online Food Ordering System allows customers to register/login, browse food items, place orders, make payments, and track order status. Restaurants manage incoming orders and administrators manage system information and reports.

Software/Tools: Draw.io, Lucidchart, Microsoft Visio, Canva, or any diagram-drawing software.

1. Data Flow Diagram – Level 0

A Level 0 DFD, also called a context diagram, shows the complete system as one process and identifies the main external entities that exchange data with the system.

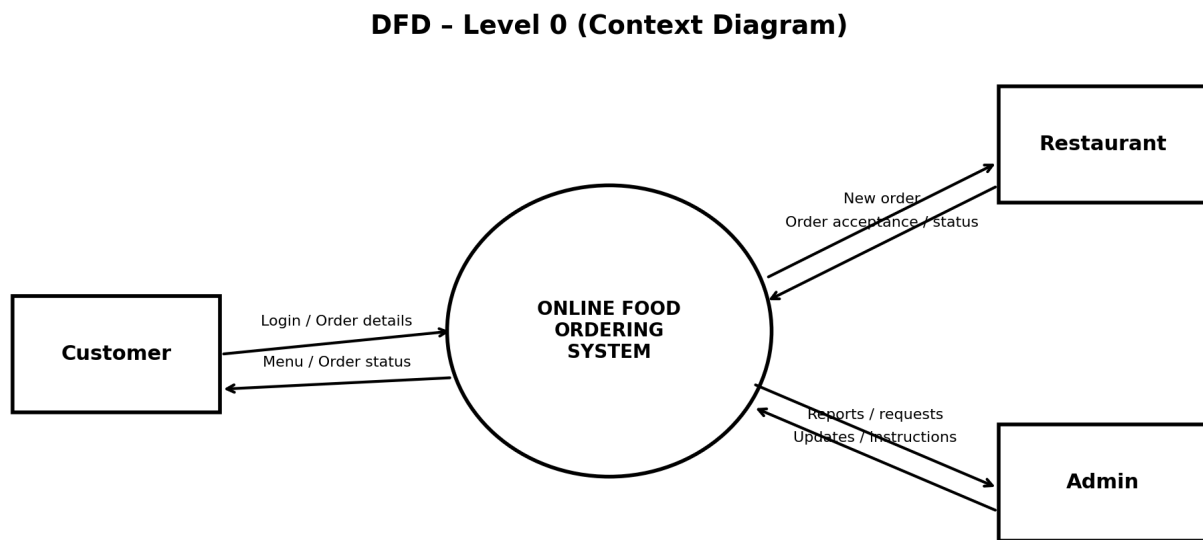


Figure 1: Context-level DFD of the Online Food Ordering System.

2. Data Flow Diagram – Level 1

The Level 1 DFD decomposes the main system into major processes such as login, menu browsing, order placement, payment, order management, and reporting. Data stores are also shown.

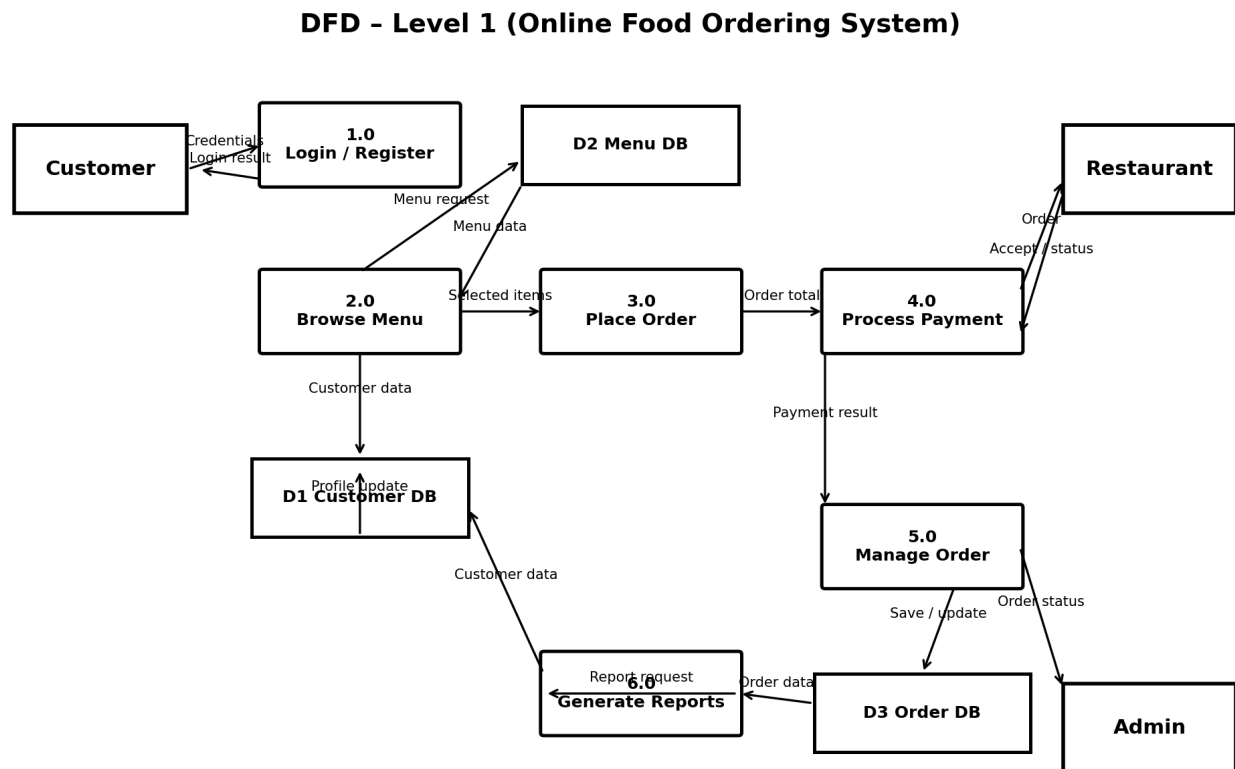


Figure 2: Level 1 DFD of the Online Food Ordering System.

3. Decision Table

A decision table represents business rules using conditions and actions. The following table decides whether an order can be confirmed.

Conditions / Actions	Rule 1	Rule 2	Rule 3	Rule 4
Customer logged in?	Yes	Yes	No	Yes
Food item available?	Yes	No	Yes	Yes
Payment successful?	Yes	-	-	No
Confirm order	Yes	No	No	No
Show payment retry	No	No	No	Yes
Ask customer to login	No	No	Yes	No
Show item unavailable	No	Yes	No	No

Interpretation: The order is confirmed only when the customer is logged in, the food is available, and payment is successful. If payment fails, the customer is asked to retry. If the customer is not logged in, the system asks for login. If the food is unavailable, the system informs the customer.

4. E-R (Entity-Relationship) Diagram

The E-R diagram represents the main entities of the system, their attributes, primary/foreign keys, and relationships.

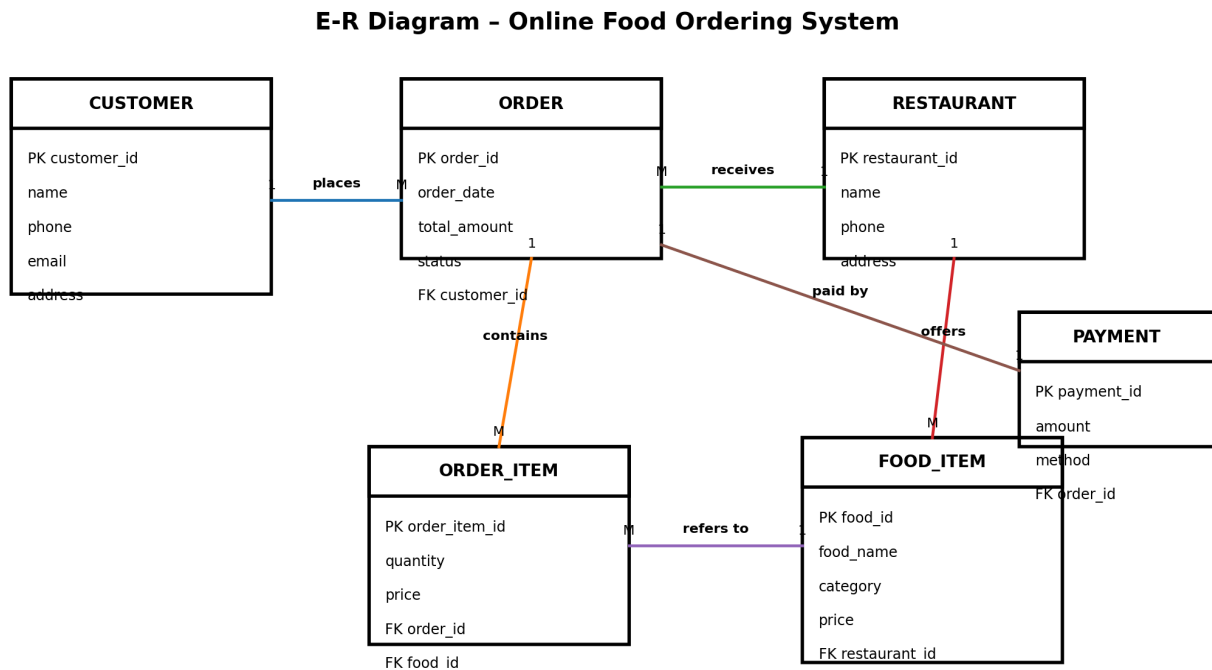


Figure 3: E-R diagram of the Online Food Ordering System.

Main Entities: Customer, Order, Order Item, Food Item, Restaurant, and Payment.

Important Relationships: A customer can place many orders; an order contains many order items; a restaurant offers many food items; an order is associated with payment; and each order item refers to a food item.

5. Conclusion

The Online Food Ordering System was analyzed using three important software engineering modelling techniques: Data Flow Diagrams, a Decision Table, and an E-R Diagram. The DFDs show how data moves through the system, the decision table represents order-confirmation rules, and the E-R diagram shows the database entities and their relationships. Students can use this sample as a reference and create similar models for their own project.